🎯 SECOND OPPORTUNITY EXAM — PRACTICE QUESTIONS

Distributed Systems (CMP3851 / S4831DD)

Based on: Your First Exam + Past Papers (2020, 2022, 2023) + Course Outline

Topics NEVER tested across ALL exams but ARE in your course outline

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PART 1: TOPICS NEVER TESTED — PRACTICE QUESTIONS

CHAPTER 1: INTRODUCTION TO DISTRIBUTED SYSTEMS

(Already tested: Definition, Characteristics, Transparency, Challenges, Examples) **Q1.1** [6 marks] Explain the three types of distributed systems and give one example of each.

- Distributed Computing Systems
- Distributed Information Systems
- Pervasive and Mobile Systems

Q1.2 [4 marks] Distinguish between cluster computing and grid computing in terms of their key characteristics.

Q1.3 [4 marks] Describe four key trends driving the development and use of distributed systems today.

Q1.4 [6 marks] Define middleware and explain its purpose in distributed systems. Give two examples of middleware.

Q1.5 [4 marks] Explain the difference between RPC (Remote Procedure Calls) and RMI (Remote Method Invocation). What disadvantage do they share that led to Message-Oriented Middleware (MOM)?

Q1.6 [4 marks] List and briefly describe four high-level features of middleware.

Q1.7 [4 marks] What are the benefits of distributed systems? List and explain four benefits.

CHAPTER 2: ARCHITECTURES

(Already tested: Architectural styles, Client-server, P2P, Tiered architectures) **Q2.1** [4 marks] Explain the two design patterns used in organizing middleware: wrappers/adapters and interceptors.

Q2.2 [6 marks] Describe the three system models used to view and design distributed systems: Physical model, Architectural model, and Fundamental model.

Q2.3 [6 marks] Explain the three generations of distributed systems, including their time periods and key characteristics.

Q2.4 [4 marks] Describe three examples of contemporary distributed systems (mobile computing, ubiquitous computing, cloud computing, or ULS systems).

Q2.5 [6 marks] What are the four key questions that architectural elements help answer in distributed systems? Explain each briefly.

Q2.6 [4 marks] Explain the three types of communication paradigms in distributed systems.

Q2.7 [4 marks] What is placement in distributed systems? Explain caching, mobile code, and mobile agents.

Q2.8 [6 marks] Describe three architectural patterns other than tiered architecture: proxy pattern, brokerage pattern, and reflection pattern.

Q2.9 [6 marks] Distinguish between synchronous and asynchronous distributed systems. What are the bounds (or lack thereof) in each?

Q2.10 [4 marks] Define latency, bandwidth, and jitter in the context of communication channel performance.

Q2.11 [4 marks] What is clock drift rate? How can computers synchronize their clocks?

Q2.12 [6 marks] Explain the Failure Model in distributed systems. Describe omission failures, arbitrary/ Byzantine failures, timing failures, and masking failures.

Q2.13 [4 marks] What are the two properties of reliable one-to-one communication? Explain validity and integrity.

Q2.14 [4 marks] Describe the Security Model in distributed systems. How can threats be resisted?

CHAPTER 3: PROCESSES

(Already tested: Threads, Multithreading, Client-server vs. P2P) **Q3.1** [6 marks] What is virtualization? Explain how it applies to distributed systems and describe the three cloud service types: IaaS, PaaS, and SaaS.

Q3.2 [4 marks] Explain the role of client processes and client software in distributed systems. How do they achieve distribution transparency?

Q3.3 [4 marks] Distinguish between iterative and concurrent servers. Also distinguish between stateless and stateful servers.

Q3.4 [6 marks] What is code migration? Explain the two main reasons for migrating code and the problem it faces in heterogeneous systems.

Q3.5 [4 marks] What is a mobile agent? How does it differ from mobile code?

CHAPTER 4: COMMUNICATION

(Already tested: Communication fundamentals, IPC, RPC/RMI, OSI layers, Sync/Async, Connection types) **Q4.1** [4 marks] List and briefly explain four standard communication protocols used in distributed systems.

Q4.2 [4 marks] What is message-oriented communication? Explain message queues and persistent asynchronous communication.

Q4.3 [4 marks] What is multicast communication? Explain one-to-many communication and its use in distributed coordination.

Q4.4 [4 marks] Explain the two types of request-reply protocols: synchronous and asynchronous.

Q4.5 [4 marks] What is indirect communication? Explain space uncoupling and time uncoupling.

CHAPTER 5: NAMING AND NAME SERVICES

(Already tested: Names, Identifiers, Addresses, DNS, URL, Naming types, Resolution)
Q5.1 [4 marks] What are naming services? Explain three key benefits of naming services.

Q5.2 [4 marks] Explain the role of names and name services in distributed systems. Why are names preferable to identifiers?

Q5.3 [4 marks] What is a name space? Explain naming context with examples.

Q5.4 [4 marks] What is binding in the context of naming? Explain naming domain, alias, and symbolic name.

Q5.5 [4 marks] What is navigation in distributed systems? How do URLs facilitate navigation?

Q5.6 [4 marks] What are location systems? Why is location transparency important in distributed systems?

CHAPTER 6: SYNCHRONIZATION AND COORDINATION

(Already tested: Clock sync, Physical/logical clocks, Happened-before, Lamport/Vector clocks, Global state, ACID, Transactions, Serializability, Concurrency, Deadlock) **Q6.1** [4 marks] Why is clock synchronization important in distributed systems? Give three reasons.

Q6.2 [4 marks] Explain the difference between external synchronization and internal synchronization of physical clocks.

Q6.3 [4 marks] Distinguish between physical time and logical time in distributed systems.

Q6.4 [4 marks] What is a global state in distributed systems? List four purposes for capturing global states.

Q6.5 [4 marks] What is serial equivalence? How does it differ from serializability?

Q6.6 [4 marks] What are conflicting operations? Give an example.

Q6.7 [6 marks] Explain the three strategies for scheduling operations in a transaction.

Q6.8 [4 marks] What are nested transactions? Give two advantages of nested transactions.

Q6.9 [4 marks] What are the two main components of distributed transaction implementations?

Q6.10 [6 marks] Distinguish between pessimistic and optimistic concurrency control. When is each approach most suitable?

Q6.11 [4 marks] What are the two types of locks? How do they differ? What is a lock manager?

Q6.12 [4 marks] List three disadvantages of using locks in transactions.

Q6.13 [6 marks] Explain three methods for deadlock prevention and handling: upgrading locks, deadlock detection, and timeouts.

Q6.14 [4 marks] What is Optimistic Concurrency Control (OCC)? How does it work?

Q6.15 [4 marks] What is timestamp ordering? Why is it advantageous for read-only transactions?

Q6.16 [6 marks] Explain the atomic commit protocol. What happens during a commit? What happens during an abort?

CHAPTER 7: FAULT TOLERANCE

(Already tested: Fault definition, Fault tolerance, Fault types, Failure models, Byzantine failures, Network partitions) **Q7.1** [6 marks] Classify faults into three categories and explain each with examples.

Q7.2 [4 marks] What is a dependable system? List and explain four dependability requirements.

Q7.3 [6 marks] Describe five types of failures in distributed systems with examples for each.

Q7.4 [6 marks] Explain the three types of redundancy used in failure handling: information redundancy, time redundancy, and physical redundancy.

Q7.5 [4 marks] What is process resilience? How is it achieved?

Q7.6 [6 marks] Explain failure masking and replication. How does a group of identical processes allow masking of faulty processes?

Q7.7 [4 marks] Describe two mechanisms for detecting process failures.

Q7.8 [4 marks] How is reliable client-server communication established? Explain acknowledgments and retry mechanisms.

Q7.9 [4 marks] What is reliable group communication? What do reliable multicast services guarantee?

Q7.10 [6 marks] Describe three recovery mechanisms used in distributed systems.

CHAPTER 8: SECURITY

(Already tested: CIA triad, Attack types, Cryptography types, Email vulnerabilities, Security mechanisms, Blockchain) **Q8.1** [4 marks] Define the four security concepts: vulnerability, threat, risk, and attack.

Q8.2 [6 marks] Describe the four categories of security threats: disclosure, deception, disruption, and usurpation.

Q8.3 [8 marks] Explain the following security threats with examples: masquerading, repudiation of origin, denial of receipt, delay, denial of service, and replaying.

Q8.4 [4 marks] Distinguish between active attacks and passive attacks. Give two examples of each.

Q8.5 [4 marks] What is a security policy? How are security policies enforced?

Q8.6 [6 marks] List and explain five security mechanisms used in distributed systems.

Q8.7 [4 marks] List six methods of authentication used in distributed systems.

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Q8.8 [4 marks] What is the difference between authorization and identification?

Q8.9 [4 marks] What is auditing? Why is it useful in security management?

Q8.10 [4 marks] Explain Access Control List (ACL) and Certificates as two approaches to access control.

Q8.11 [4 marks] List five security tools used in distributed systems.

Q8.12 [6 marks] Describe three security design issues: simplicity, layering, and control.

Q8.13 [4 marks] What is a cryptographic key? Explain shared secret keys and public/private keys.

Q8.14 [4 marks] Give one example each of an asymmetric cryptosystem and a symmetric cryptosystem.

Q8.15 [4 marks] What are secure channels? Name two widely-used components that support secure communication.

Q8.16 [6 marks] What is a certificate? What are its requirements? Give an example.

Q8.17 [4 marks] What is a digital signature? How is it implemented?

Q8.18 [6 marks] What is Public Key Infrastructure (PKI)? How does it work?

Q8.19 [4 marks] What is key management? What is authorization management?

PART 2: CROSS-CHAPTER INTEGRATED QUESTIONS (High Probability for Retake)

These questions combine multiple untested topics and mirror the style of past exams.

QI.1 [10 marks] Mr. Haiduwa's course notes discuss three generations of distributed systems and three types of distributed systems. (a) Describe the three generations of distributed systems. [6 marks] (b) Explain how cloud computing fits into both the "types" and "generations" frameworks. [4 marks]

QI.2 [10 marks] (a) Define middleware and explain its role in distributed systems. [4 marks] (b) Describe two types of communication middleware (RPC and RMI) and explain the disadvantage that led to Message-Oriented Middleware. [6 marks]

QI.3 [10 marks] (a) Explain the three fundamental models of distributed systems. [6 marks] (b) How does the interaction model differ between synchronous and asynchronous distributed systems? [4 marks]

QI.4 [10 marks] (a) What is virtualization? Explain IaaS, PaaS, and SaaS. [6 marks] (b) How does virtualization solve the problem of code migration in heterogeneous systems? [4 marks]

QI.5 [10 marks] (a) Distinguish between stateless and stateful servers. Give one advantage of each. [4 marks] (b) Explain code migration and give two reasons why it is used. [6 marks]

QI.6 [10 marks] (a) What is message-oriented communication? How does it differ from remote invocation? [4 marks] (b) Explain multicast communication and give two situations where it is useful. [6 marks]

QI.7 [10 marks] (a) Explain binding, naming domain, and alias in the context of naming services. [6 marks] (b) What is navigation? How do URLs facilitate navigation? [4 marks]

QI.8 [10 marks] (a) Distinguish between pessimistic and optimistic concurrency control. [4 marks] (b) Explain timestamp ordering and why it cannot suffer from deadlocks. [6 marks]

QI.9 [10 marks] (a) What is the atomic commit protocol? Explain what happens during commit and abort. [6 marks] (b) What are nested transactions? Give two advantages. [4 marks]

QI.10 [10 marks] (a) Explain the three types of redundancy used in failure handling with examples. [6 marks] (b) Describe two mechanisms for fault detection in distributed systems. [4 marks]

QI.11 [10 marks] (a) What is process resilience? How is it achieved through replication and checkpointing? [6 marks] (b) Explain reliable group communication and what it guarantees. [4 marks]

QI.12 [10 marks] (a) Describe the four categories of security threats with one example each. [8 marks] (b) What is the difference between a security policy and a security mechanism? [2 marks]

QI.13 [10 marks] (a) Explain access control using ACL and certificates. [6 marks] (b) What is a digital signature and how is it implemented? [4 marks]

QI.14 [10 marks] (a) What is PKI and how does it work? [6 marks] (b) Explain secure channels and name two widely-used protocols. [4 marks]

QI.15 [10 marks] (a) List and explain five security mechanisms. [5 marks] (b) Describe three security design issues in distributed systems. [5 marks]

PART 3: QUICK REFERENCE — TOPICS TESTED vs. NEVER TESTED

Chapter	Section	Tested?	Priority for Retake
1.1	Definition of DS	✅ Yes (all exams)	Low
1.2	Characteristics	✅ Yes (all exams)	Low
1.3	Design goals & challenges	✅ Yes (all exams)	Low
1.4	Types of distributed systems	❌ NEVER	🔴 HIGH
1.5	Types of transparency	✅ Yes (all exams)	Low
1.6	Examples of DS	✅ Yes (all exams)	Low
1.7	Middleware	❌ NEVER	🔴 HIGH
2.1	Architectural styles	✅ Yes (all exams)	Low
2.2	System architectures	✅ Yes (all exams)	Low
2.3	Architecture examples	✅ Yes (all exams)	Low
2.4	Middleware organization	❌ NEVER	🔴 HIGH
2.5	System models	❌ NEVER	🔴 HIGH
3.1	Threads/ multithreading	✅ Yes (all exams)	Low
3.2	Virtualization	❌ NEVER	🔴 HIGH
3.3	Clients	❌ NEVER	🟡 MEDIUM
3.4	Servers	❌ NEVER	🟡 MEDIUM
3.5	Client-server vs. P2P	✅ Yes (all exams)	Low
3.6	Code migration	❌ NEVER	🔴 HIGH

Table 1 – continued

Chapter	Section	Tested?	Priority for Retake
4.1	Communication fundamentals	✅ Yes (all exams)	Low
4.2	IPC	✅ Yes (all exams)	Low
4.3	RPC/RMI	✅ Yes (all exams)	Low
4.4	Client-server communication	✅ Yes (all exams)	Low
4.5	Message-oriented communication	❌ NEVER	🔴 HIGH
4.6	Multicast communication	❌ NEVER	🔴 HIGH
5.1	Names, identifiers, addresses	✅ Yes (all exams)	Low
5.2	Flat naming	✅ Yes (all exams)	Low
5.3	Structured naming	✅ Yes (all exams)	Low
5.4	Attribute-based naming	✅ Yes (all exams)	Low
5.5	Location systems	❌ NEVER	🟡 MEDIUM
6.1	Synchronization & coordination	✅ Yes (all exams)	Low
6.2	Clock synchronization	✅ Yes (all exams)	Low
6.3	Physical & Logical clocks	Partially tested	🟡 MEDIUM
6.4	Transactions	✅ Yes (all exams)	Low
6.5	Concurrency control	✅ Yes (all exams)	Low
6.6	Deadlock handling	Partially tested	🟡 MEDIUM
7.1	Fault tolerance	✅ Yes (all exams)	Low
7.2	Fault models	✅ Yes (all exams)	Low
7.3	Fault detection	❌ NEVER	🔴 HIGH
7.4	Process resilience	❌ NEVER	🔴 HIGH
7.5	Reliable client-server communication	❌ NEVER	🟡 MEDIUM
7.6	Recovery mechanisms	❌ NEVER	🔴 HIGH
8.1	Intro to security	✅ Yes (all exams)	Low
8.2	Security concepts (threats, policies, mechanisms)	Partially tested	🟡 MEDIUM
8.3	Security design issues	❌ NEVER	🟡 MEDIUM
8.4	Secure channels	❌ NEVER	🔴 HIGH
8.5	Access control	❌ NEVER	🔴 HIGH

Table 1 – continued

Chapter	Section	Tested?	Priority for Retake
8.6	Secure naming	❌ NEVER	🔴 HIGH
8.7	Security management	❌ NEVER	🔴 HIGH

PART 4: TOP 20 MUST-KNOW ANSWERS (Priority Study List)

1. Types of Distributed Systems [6 marks]

Distributed Computing Systems: High-performance computing. Sub-types:

- Cluster: Similar workstations, high-speed LAN, homogeneous (same OS)
- Grid: Federation of systems, different admin domains, heterogeneous, resource sharing via virtual organizations
- Cloud: Virtualized resources, charged per-use

Distributed Information Systems: Database applications using distributed transactions (transaction processing, enterprise application integration)

Pervasive & Mobile Systems: Small, battery-powered, mobile, wireless. Includes ubiquitous computing, home systems, e-healthcare, sensor networks

2. Middleware [6 marks]

- Layer of software between applications and OS
- Purpose: Higher-level programming abstraction, mask heterogeneity
- RPC: Local procedure call packaged as message
- RMI: Calls to remote objects
- Disadvantage: Tight coupling (both must be running) → led to MOM
- Examples: Java RMI, JBoss, Websphere, Apache

3. Three Generations of DS [6 marks]

- Early (1970s-80s): 10-100 nodes, LAN, limited Internet
- Internet-scale (1990s): Network of networks, extensible nodes
- Contemporary: Mobile, ubiquitous, cloud, ULS systems

4. System Models [6 marks]

- Physical: Hardware composition, computers, networks
- Architectural: Computational and communication tasks
- Fundamental: Interaction, Failure, Security models

5. Synchronous vs. Asynchronous [6 marks]

- Synchronous: Known bounds on execution time, message delay, clock drift
- Asynchronous: No bounds (models the Internet)

6. Virtualization & Cloud Services [6 marks]

- Virtualization: Extending/replacing interface to mimic another system
- IaaS: Infrastructure (basic hardware)
- PaaS: Platform (system-level services)
- SaaS: Software (actual applications)

7. Code Migration [6 marks]

- Passing program codes between machines
- Reasons: Performance (load balancing), Flexibility (partition application)
- Problem in heterogeneous systems: Compatibility → solved by VMs

8. Stateless vs. Stateful Servers [4 marks]

- Stateless: No client state info; forgets client after request
- Stateful: Maintains persistent client info between operations

9. Message-Oriented Communication [4 marks]

- Deals with persistence and synchronicity
- Message queues: Producer sends to queue, consumer receives
- Persistent async communication: Messages stored until delivered

10. Multicast Communication [4 marks]

- One sender to multiple receivers
- Used in distributed coordination, flooding messages

11. Naming Services Benefits [4 marks]

- Resource localization, Uniform naming, Device independence

12. Binding & Navigation [4 marks]

- Binding: Associating name to object
- Naming domain: Authority assigning names
- Alias: Alternative name (e.g., www, ftp)
- Navigation: Chaining naming services to resolve names
- URLs: Decompose into protocol, endpoint, path

13. Pessimistic vs. Optimistic Concurrency Control [6 marks]

- Pessimistic: Lock before use; faster if conflicts frequent
- Optimistic: No locking; validate at commit; faster if conflicts rare

14. Timestamp Ordering [4 marks]

- Orders transactions by start time
- Cannot suffer from deadlocks
- Good for read-only transactions
- Multi-version: Particularly effective

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15. Atomic Commit Protocol [6 marks]

- Commit: All changes permanent; future transactions see results
- Abort: Undoes modifications; no effects visible to future transactions

16. Nested Transactions [4 marks]

- Transactions from sub-transactions
- Advantages: Concurrent execution in separate servers, independent recovery

17. Three Types of Redundancy [6 marks]

- Information: Extra bits (Hamming code)
- Time: Perform action again (retry transaction)
- Physical: Extra equipment/processes

18. Fault Detection [4 marks]

- Heartbeat: "Are you alive?" messages (pinging)
- Monitoring services: Regular failure checks

19. Process Resilience [4 marks]

- Protection against process failures
- Replication: Identical process groups
- Checkpointing: Save state for recovery

20. PKI & Digital Signatures [6 marks]

- PKI: Knows public key belongs to specific user
- Builds on asymmetric encryption
- Private key: Known only to entity
- Public key: In X.509 certificate
- Digital signature: Uniquely tied to content; verified via RSA

Good luck! Focus on the 🔴 HIGH priority topics first — they have never been tested but are major course concepts.

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